

Active Directory Certificate Services (AD CS) – Practical

Task 1: Install Active Directory Certificate Services

Question:

Open Server Manager on your Windows Server VM and install the Active Directory Certificate Services (AD CS) role, selecting only the Certification Authority role service.

Steps:

1. Log in to **Windows Server** as Administrator.
 2. Open **Server Manager**.
 3. Click **Manage → Add Roles and Features**.
 4. Click **Next** on the Before You Begin page.
 5. Select **Role-based or feature-based installation → Next**.
 6. Select your Windows Server from the server pool → **Next**.
 7. Select **Active Directory Certificate Services**.
 8. Click **Add Features** when prompted.
 9. Click **Next** until you reach **Role Services**.
 10. Select only **Certification Authority**.
 11. Click **Next → Install**.
 12. Wait for the installation to complete.
 13. Click **Close**.
 14. In Server Manager, click the notification flag and select **Configure Active Directory Certificate Services on the destination server**.
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Task 2: Configure a Standalone Root CA

Question:

Configure a new Standalone Root CA using the AD CS Configuration Wizard and name the CA after your favorite IPL team, such as GujaratTitansRootCA.

Steps:

1. In the AD CS Configuration wizard, select **Certification Authority**.

2. Click **Next**.
 3. For Setup Type, select **Standalone CA**.
 4. Click **Next**.
 5. For CA Type, select **Root CA**.
 6. Click **Next**.
 7. Select **Create a new private key**.
 8. Click **Next**.
 9. Keep the default cryptographic settings unless your lab requires different settings.
 10. In **Common name for this CA**, enter:
GujaratTitansRootCA
 11. Click **Next**.
 12. Set the certificate validity period according to the lab requirement. For example, **5 years**.
 13. Select the default database and log locations.
 14. Click **Next → Configure**.
 15. Wait until the configuration completes successfully.
 16. Click **Close**.
 17. Open **Certification Authority** from **Server Manager → Tools**.
 18. Verify that GujaratTitansRootCA is displayed.
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Task 3: Create an App Developers Certificate and Export It

Question:

Create a certificate for **App Developers** that allows code signing, issue a certificate from your CA, and export it as a .cer file.

Steps:

1. Open **Certification Authority** from **Server Manager → Tools**.
2. Open your CA, for example **GujaratTitansRootCA**.
3. Open **Certificate Templates**.

4. Right-click **Certificate Templates** → **Manage**.
 5. Locate a suitable **Code Signing** certificate template.
 6. If your lab requires a custom **App Developers** template, duplicate the suitable template and name the new template:
App Developers.
 7. Open the template **Properties**.
 8. Under **Extensions**, verify that **Code Signing** is included in the Application Policies/EKU.
 9. Apply the settings.
 10. Return to the Certification Authority console.
 11. Right-click **Certificate Templates** → **New** → **Certificate Template to Issue**.
 12. Select **App Developers** and click **OK**.
 13. On the client/server, open mmc.
 14. Add the **Certificates** snap-in for **Current User**.
 15. Go to **Personal** → **Certificates**.
 16. Right-click and select **All Tasks** → **Request New Certificate**.
 17. Select the **App Developers** certificate.
 18. Click **Enroll**.
 19. After successful enrollment, open the issued certificate.
 20. Select **Details** and verify that **Code Signing** is listed in the certificate's intended usage.
 21. Right-click the certificate → **All Tasks** → **Export**.
 22. Select **No, do not export the private key**.
 23. Select **Base-64 encoded X.509 (.CER)** or **DER encoded binary X.509 (.CER)**.
 24. Save the file as:
AppDevelopers.cer
 25. Finish the export wizard.
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Task 4: Create and Issue a Certificate for ZomatoAPI

Question:

Generate a certificate request (CSR) for an imaginary ZomatoAPI server, submit it to your CA, approve it, and install the issued certificate.

Step 1 – Create the CSR

1. Log in to the **ZomatoAPI** server.
2. Open **Notepad** as Administrator.
3. Create a file named:
ZomatoAPI.inf
4. Add the following configuration:

[Version]

Signature="\$Windows NT\$"

[NewRequest]

Subject = "CN=ZomatoAPI"

KeyLength = 2048

HashAlgorithm = sha256

KeyAlgorithm = RSA

MachineKeySet = TRUE

Exportable = TRUE

RequestType = PKCS10

❓ Save the file as ZomatoAPI.inf.

❓ Open **Command Prompt as Administrator**.

❓ Run:

```
certreq -new ZomatoAPI.inf ZomatoAPI.req
```

8. Verify that ZomatoAPI.req has been created.

Step 2 – Submit the CSR to the CA

9. Copy ZomatoAPI.req to the CA server.
10. On the CA server, open **Certification Authority**.
11. Right-click the CA name.
12. Select **All Tasks** → **Submit new request**.
13. Select ZomatoAPI.req.
14. Click **Open**.
15. Open **Pending Requests**.
16. Locate the ZomatoAPI request.

Step 3 – Approve the Certificate

17. Right-click the pending request.
18. Select **All Tasks** → **Issue**.
19. The request will move from **Pending Requests** to **Issued Certificates**.
20. Open **Issued Certificates** and verify that the ZomatoAPI certificate has been issued.

Step 4 – Install the Certificate

21. Right-click the issued certificate.
22. Select **Open** and go to **Details**.
23. Use **Copy to File** to export the issued certificate as:
ZomatoAPI.cer
24. Copy ZomatoAPI.cer back to the ZomatoAPI server.
25. Open **Command Prompt as Administrator**.
26. Run:

certreq -accept ZomatoAPI.cer

Verify the installed certificate using:

certlm.msc

? Go to **Personal** → **Certificates**.

❓ Verify that the **ZomatoAPI** certificate is installed.

Task 5: List All Issued Certificates Using PowerShell

Question:

Use PowerShell to list all certificates issued by your CA and take a screenshot of the output.

Steps:

1. Open **PowerShell as Administrator** on the CA server.
2. Import the AD CS administration module:

Import-Module ADCSAdministration

3. Run the following script:

```
$CA = Get-CertificationAuthority
```

```
Get-IssuedRequest -CertificationAuthority $CA |
```

```
Select-Object RequestID, CommonName, RequesterName, NotBefore, NotAfter,  
CertificateTemplate |
```

```
Format-Table -AutoSize
```

4. Press **Enter**.

5. Verify that the issued certificate requests are displayed.

6. Check for entries such as **ZomatoAPI** and other issued certificates.

7. Take a screenshot of the PowerShell output.

8. Add the screenshot to your practical report.